ECE 278C **IMAGING SYSTEMS** **Fall 2025**

Assignment 8: Short Baseline Geolocation Systems (Approximation techniques)

Due: Monday, December 8

The objective of this assignment is to modify the software program developed in the previous assignment for the geolocation of a single target in the active-sensing mode using 2 approximation techniques.

Method 1: (Monostatic model)

Approximate the bistatic short-baseline sensing system as a pair of active monostatic units located at $x = \pm D/2$, with the data sequences $g_1(n)$ and $g_2(n)$ respectively. And then construct the target profile as the intersection of two bands of circles. An N-frame video can be constructed to illustrate the convergence of the geolocation task.

Method 2: (Range and bearing-angle model, polar format)

Approximate and convert the exercise into a range-estimation problem with the sequence $g_1(n)\, g_2(n)$

$$g_1(n)\, g_2(n) \;=\; A^2 \exp(\, j2\pi f_n\, (\tau_1 + \tau_2\,))\;=\; A^2 \exp(\, j2\pi\, (2r_0 + r_1 + r_2)/\lambda_n\,)$$

$$\approx\; A^2 \exp(\, j2\pi\, (4r_0)/\lambda_n\,)$$

Perform the range-estimation task in the polar (θ, r) coordinate system with the sequence $g_1(n)\, g_2(n)$. The range profile $R(\theta, r) = R(r)$ is in the form of a band of horizontal lines.

Similarly, convert the task into a bearing-angle estimation problem with the sequence $g_1(n)\, g_2{}^*(n)$

$$g_1(n)\, g_2{}^*(n) \;=\; A^2 \exp(\, j2\pi f_n\, (\tau_1 - \tau_2\,))\;=\; A^2 \exp(\, j2\pi\, (r_1 - r_2)/\lambda_n\,)$$

Subsequently, model it as a bearing-angle estimation problem with the sequence $g_1(n)\, g_2{}^*(n)$. The bearing-angle profile $q(\theta, \mathrm{r}) = q(\theta)$ is a band of vertical lines.

An N-frame video can be constructed to illustrate the convergence of the geolocation task in the polar coordinate system.

Report:

a. Cover sheet
b. Final images from the two methods and compare against the result from Assignment 7.
c. Video sequences from the two methods and compare against the result from Assignment 7.
d. The quality of the geolocation techniques is defined by (a) accuracy of the target location and (b) the resolution of the target profile. Discuss the effectiveness of these approximation techniques with respect to (1) separation distance *2D*, (2) bandwidth, (3) number of the frequency steps, (4) range distance and bearing angle of the target location.
e. Matlab code
f. Summary